XIAMEN UNIVERSITY MALAYSIA

FINAL EXAMINATION

Course Code:	EEE301			
Course Name:	Signals and Systems			
Question Paper Setter:	Tee Wei Hown			
Academic Session:	2024/04	Question Paper:	A ☑	B ☐
Total No. of Pages:	4 (exclude cover page)	Time Allocated:	2 hours	
Additional Materials:	APPENDIX A			
Apparatus Allowed:	Scientific Calculator			

INSTRUCTIONS TO CANDIDATES

1. This paper consists of FOUR (4) questions. Answer ALL questions.
2. APPENDIX A is attached to this question book for your reference.
3. Read the above information carefully to ensure you have the correct and complete question paper.
4. Please follow the requirement of each section and write down all the corresponding answers on the answer book provided.
5. Candidates are not allowed to leave the exam room during the first 60 minutes and last 15 minutes of the exam.
6. Candidates are required to hand in both question paper and answer book to the invigilator before leaving the examination room (including all blank or used papers, if applicable).

DO NOT TURN OVER THIS PAGE UNTIL INSTRUCTED TO DO SO.

(Student ID: _____ Full Name: _____)

QUESTION 1 (30 Marks) [CLO1-PLO1:C2]

Consider the cascade interconnection of three causal LTI systems as Figure 1 (a) below:

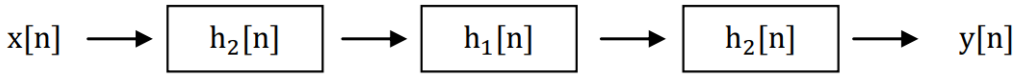


Figure 1 (a)

The impulse response of $h_2[n]$ is

$$h_2[n] = u[n] - u[n-2]$$

and the impulse response of the overall system is shown in Figure 1 (b).

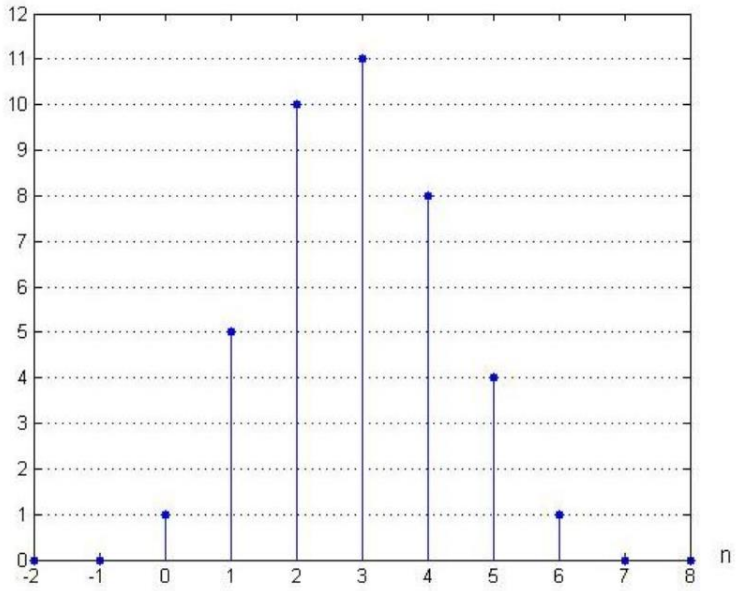


Figure 1 (b)

a.) Find plot the impulse response of $h_1[n]$.

[CLO1-PLO1:C2][10 Marks]

b.) Find and plot the response of the overall system to the input $x[n]$.

$$x[n] = \delta[n] - \delta[n-1]$$

[CLO1-PLO1:C2][10 Marks]

c.) A system's impulse response is $h(t) = \frac{u(t-1)}{t^2}$ and $h(t) = \frac{u(t-1)}{t}$. Justify if the system is BIBO stable or not stable.

[CLO1-PLO1:C2][10 Marks]

QUESTION 2 (20 Marks) [CLO2-PLO2:C4]

LTI systems can be represented by using CTFT in time and frequency domain. Figure 2 shows a RC circuit excited by a voltage source, $v_s(t)$ with $R_1 = 2k\Omega$, $R_2 = 4k\Omega$ and $C = 0.25mF$.

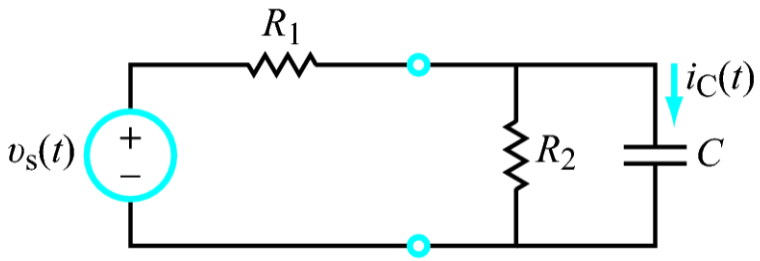


Figure 2

Determine the voltage across the capacitor, $v_c(t)$ if

a.) The voltage of the source $v_s(t) = 10u(t)$.

[CLO2-PLO2:C4][10 Marks]

b.) The voltage of the source $v_s(t) = 10e^{-2t}u(t)$

[CLO2-PLO2:C4][10 Marks]

QUESTION 3 (20 Marks) [CLO2-PLO2:C4]

Answer the following questions accordingly.

a.) Determine the Laplace transform of $x_1(t)$ and $x_2(t)$ if

$$x_1(t) = [\sin \omega_o t]u(t)$$
$$x_2(t) = (t - T)u(t - T)$$

[CLO2-PLO2:C4][12 Marks]

b.) Determine the inverse Laplace transform of

$$X(s) = \frac{4s^2 - 5s - 10}{(s + 2)^3}$$

[CLO2-PLO2:C4][8 Marks]

QUESTION 4 (30 Marks) [CLO3-PLO2:C3]

Build the transfer functions of these Butterworth filters:

a.) Second-order high pass with a cutoff frequency of 20 kHz and a passband gain of 5. The transfer function of the normalized filter is

$$H(s) = \frac{K}{(s+0.707+j0.707)(s+0.707-j0.707)} \text{ where } s = \frac{\omega c}{s}$$

[CLO3-PLO2:C3][8 Marks]

b.) Third-order bandpass with a center frequency of 5 kHz, a -3 dB bandwidth of 500 Hz and a passband gain of 1. The transfer function of the normalized filter is

$$H(s) = \frac{K}{(s+1)\left(s-e^{j\frac{2\pi}{3}}\right)\left(s-e^{j\frac{2\pi}{3}}\right)} \text{ where } s = \frac{s^2+\omega L\omega H}{s(\omega H-\omega L)}$$

[CLO3-PLO2:C3][10 Marks]

c.) Fourth-order bandstop with a center frequency of 10 MHz, a -3 dB bandwidth of 50 kHz and a passband gain of 1. The transfer function of the normalized filter is

$$H(s) = \frac{1}{s^4+2.6131s^3+3.4142s^2+2.6131s+1} \text{ where } s = \frac{s(\omega H-\omega L)}{s^2+\omega L\omega H}$$

[CLO3-PLO2:C3][12 Marks]

- END OF PAPER -

APPENDIX A

Fourier Transform pairs (CT signal)

CT Signals		Time Domain	Frequency Domain
1.	Impulse	$\delta(t)$	1
2.	Constant	1	$2\pi\delta(\omega)$
3.	Unit step function	$u(t)$	$\pi\delta(\omega)+\frac{1}{j\omega}$
4.	Causal decaying exponential function	$e^{-at}u(t), a>0$	$\frac{1}{a+j\omega}$
5.	Two-sided decaying exponential function	$e^{-a\lvert t\rvert}, a>0$	$\frac{2a}{a^2+\omega^2}$
6.	First-order time-rising causal decaying exponential function	$te^{-at}u(t), a>0$	$\frac{1}{(a+j\omega)^2}$
7.	Complex exponential	$e^{j\omega_o t}$	$2\pi\delta(\omega-\omega_o)$
8.	Periodic cosine function	$\cos(\omega_o t)$	$\pi[\delta(\omega-\omega_o)+\delta(\omega+\omega_o)]$
9.	Periodic sine function	$\sin(\omega_o t)$	$\frac{\pi}{j}[\delta(\omega-\omega_o)-\delta(\omega+\omega_o)]$
10.	Causal cosine function	$\cos(\omega_o t)\,u(t)$	$\frac{\pi}{2}[\delta(\omega-\omega_o)+\delta(\omega+\omega_o)]+\frac{j\omega}{{\omega_o}^2-\omega^2}$
11.	Causal sine function	$\sin(\omega_o t)\,u(t)$	$\frac{\pi}{2j}[\delta(\omega-\omega_o)-\delta(\omega+\omega_o)]+\frac{\omega_o}{{\omega_o}^2-\omega^2}$

Fourier and Laplace Transform

Formulation	Time Domain	Frequency Domain
Fourier Transform	$x(t)=\frac{1}{2\pi}\int_{-\infty}^{\infty}X(\omega)\,e^{j\omega t}d\omega$	$X(\omega)=\int_{-\infty}^{\infty}x(t)\,e^{-j\omega t}dt$
Laplace Transform	$x(t)=\frac{1}{2\pi}\int_{\sigma-j\infty}^{\sigma+j\infty}X(s)\,e^{st}ds$	$X(j\omega)=\int_{-\infty}^{\infty}x(t)\,e^{-j\omega t}dt$ $X(s)=\int_{-\infty}^{\infty}x(t)\,e^{-st}dt$

Laplace Transform pairs (CT signal)

CT Signals		Time Domain	Transform	ROC
1.	Impulse	$\delta(t)$	1	All s
2.	Unit step function	$u(t)$	$\frac{1}{s}$	$\mathbb{R}\{s\} > 0$
3.	Unit step function	$-u(-t)$	$\frac{1}{s}$	$\mathbb{R}\{s\} < 0$
4.	Exponential function	$e^{-at}u(t)$	$\frac{1}{s+a}$	$\mathbb{R}\{s\} > -a$
5.	Exponential function	$-e^{-at}u(-t)$	$\frac{1}{s+a}$	$\mathbb{R}\{s\} < -a$
6.	N^{th}-order time-rising causal decaying exponential function	$\frac{t^{n-1}}{(n-1)!}e^{-at}u(t)$	$\frac{1}{(s+a)^n}$	$\mathbb{R}\{s\} > -a$
7.	Impulse property	$\delta(t-T)$	e^{-sT}	All s
8.	Causal cosine function	$\cos(\omega_o t)\,u(t)$	$\frac{s}{s^2 - {\omega_o}^2 s}$	$\mathbb{R}\{s\} > 0$
9.	Causal sine function	$\sin(\omega_o t)\,u(t)$	$\frac{\omega_o}{s^2 - {\omega_o}^2 s}$	$\mathbb{R}\{s\} > 0$

Euler's Relation

$e^{jy} = \cos(y) + j\sin(y)$	$e^{-jy} = \cos(y) - j\sin(y)$
$\cos(y) = \frac{1}{2}(e^{jy} + e^{-jy})$	$\sin(y) = \frac{1}{2j}(e^{jy} - e^{-jy})$
$e^{a+b} = e^a e^b$	$Ae^{j\theta} = A(\cos\theta + j\sin\theta)$

Butterworth Polynomial Factorization

N	$D(s)$
1	$(s+1)$
2	$(s^2+1.414s+1)$
3	$(s+1)(s^2+s+1)$
4	$(s^2+0.7654s+1)(s^2+1.8478s+1)$
5	$(s+1)(s^2+0.6810s+1)(s^2+1.6810s+1)$
6	$(s^2+0.5176s+1)(s^2+1.4142s+1)(s^2+1.9319s+1)$
7	$(s+1)(s^2+0.4450s+1)(s^2+1.2470s+1)(s^2+1.8019s+1)$
8	$(s^2+0.3902s+1)(s^2+1.1111s+1)(s^2+1.6629s+1)(s^2+1.9616s+1)$
9	$(s+1)(s^2+0.3473s+1)(s^2+s+1)(s^2+1.5321s+1)(s^2+1.8794s+1)$
10	$(s^2+0.3129s+1)(s^2+0.9080s+1)(s^2+1.4142s+1)(s^2+1.7820s+1)(s^2+1.9754s+1)$

Filters Frequency Transformation

Filter Type	Transformation
Lowpass	$s=\frac{s}{\omega c}, \omega c=2\pi f_o$
Highpass	$s=\frac{\omega c}{s}, \omega c=2\pi f_o$
Bandpass	$s=\frac{s^2+\omega L\omega H}{s(\omega H-\omega L)}, \omega L=2\pi(f_o-\frac{f_b}{2}), \omega H=2\pi(f_o+\frac{f_b}{2})$
Bandstop	$s=\frac{s(\omega H-\omega L)}{s^2+\omega L\omega H}, \omega L=2\pi(f_o-\frac{f_b}{2}), \omega H=2\pi(f_o+\frac{f_b}{2})$